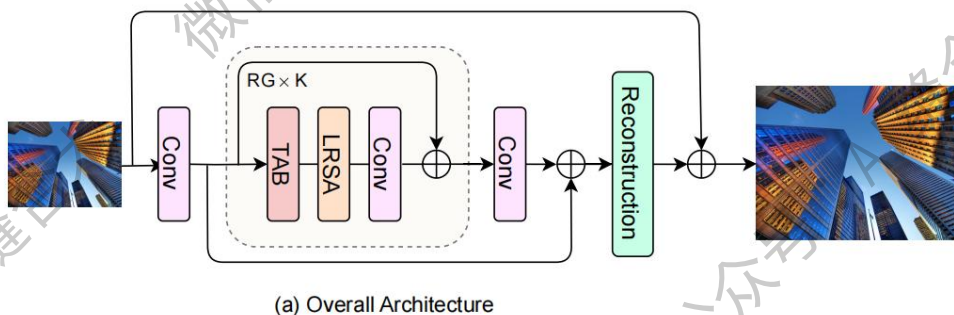
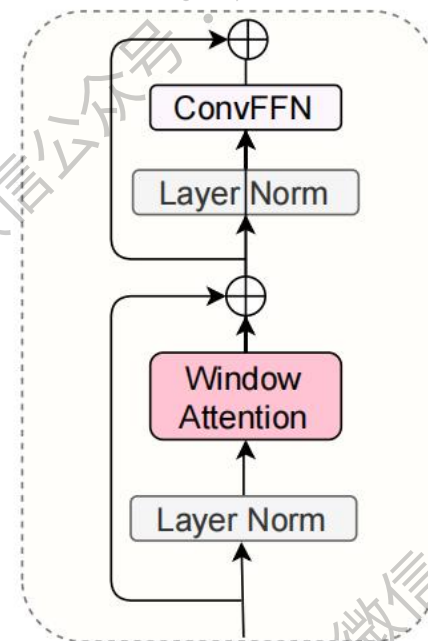


论文信息



(a) Overall Architecture



(c) Local Region Self-Attention (LRSA)

论文题目： ATANet: Efficient Content-Aware Token Aggregation for Lightweight Image Super-Resolution

中文题目： CATANet: 轻量级图像超分辨率的高效内容感知标记聚合

论文链接： <https://arxiv.org/pdf/2503.06896>

官方 github： <https://github.com/EquationWalker/CATANet/tree/main>

所属机构： 南京大学软件新技术国家重点实验室

核心速览：本文提出了一种名为 CATANet 的轻量级图像超分辨率网络，该网络通过内容感知的标记聚合模块有效捕获长距离依赖关系，同时保持高效的推理速度。

局部区域自注意力 (LRSA) : 通过使用重叠的补丁来增强局部区域特征之间的交互。

```

LRSA(
  (layer): ModuleList(
    (0): PreNorm(
      (norm): LayerNorm((32,), eps=1e-05, elementwise_affine=True)
      (fn): Attention(
        (to_q): Linear(in_features=32, out_features=36, bias=False)
        (to_k): Linear(in_features=32, out_features=36, bias=False)
        (to_v): Linear(in_features=32, out_features=32, bias=False)
        (proj): Linear(in_features=32, out_features=32, bias=False)
      )
    )
    (1): PreNorm(
      (norm): LayerNorm((32,), eps=1e-05, elementwise_affine=True)
      (fn): ConvFFN(
        (fc1): Linear(in_features=32, out_features=96, bias=True)
        (act): GELU(approximate='none')
        (dwconv): dwconv(
          (depthwise_conv): Sequential(
            (0): Conv2d(96, 96, kernel_size=(5, 5), stride=(1, 1), padding=(2, 2), groups=96)
            (1): GELU(approximate='none')
          )
        )
        (fc2): Linear(in_features=96, out_features=32, bias=True)
      )
    )
  )
)

```

微信公众号: AI缝合术!

Input shape: torch.Size([1, 32, 256, 256])

Output shape: torch.Size([1, 32, 256, 256])